

Chapter 10: Mechanics of Options Markets

Part 2 — Sections 10.4 to 10.6

Learning Objectives

After studying this part, you should be able to:

- Understand how stock options are specified by exchanges.
 - Understand expiration cycles and strike price determination.
 - Distinguish between option class and option series.
 - Understand in-the-money, at-the-money, and out-of-the-money options.
 - Understand intrinsic value and time value.
 - Learn about FLEX options and other nonstandard option products.
 - Understand stock split and dividend adjustments.
 - Understand market makers and the mechanics of option trading.
 - Understand open interest and offsetting transactions.
 - Understand commissions and their effect on profitability.
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10.4 Specification of Stock Options

A stock option contract must have standardized terms.

The exchange specifies:

- Contract size
- Expiration date
- Strike price
- Dividend adjustments
- Position limits
- Exercise procedures

Standardization improves liquidity and makes trading easier.

Standard Contract Size

In the United States:

$$1 \text{ Option Contract} = 100 \text{ Shares}$$

This standardization allows contracts to be traded efficiently on exchanges.

Expiration Dates

An option is usually identified by its expiration month.

Example:

IBM January Call

means a call option on IBM expiring in January.

Official Expiration Date

The official expiration date is:

The Saturday immediately following the third Friday of the expiration month.

However:

- Trading stops on the third Friday.
 - Investors usually submit exercise instructions before market close on Friday.
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Expiration Cycles

Stock options are assigned to one of three cycles.

January Cycle

Months:

- January
- April
- July
- October

February Cycle

Months:

- February
- May
- August
- November

March Cycle

Months:

- March
- June
- September
- December

Example: IBM (January Cycle)

Beginning of January:

Available expirations:

- January
- February

- April
- July

End of January:

Available expirations:

- February
- March
- April
- July

As one contract expires, another longer-dated contract is introduced.

LEAPS

LEAPS stands for:

Long-Term Equity Anticipation Securities

Characteristics:

- Long-dated options
 - Expiration up to 39 months away
 - Useful for long-term speculation and hedging
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Strike Prices

Exchanges choose strike prices at regular intervals.

Typical Strike Intervals

Stock Price	Strike Interval
5–25	\$2.50
25–200	\$5

Stock Price	Strike Interval
Above \$200	\$10

Example

If stock price:

$$S = 84$$

Initial strikes may be:

- 80
- 85
- 90

If stock rises above 90:

New strike introduced:

- 95

If stock falls below 80:

New strike introduced:

- 75

This ensures strike prices remain close to market prices.

Option Terminology

Many contracts may exist on the same stock.

Suppose:

- 4 expiration dates
- 5 strike prices
- Calls and puts available

Total contracts:

$$4 \times 5 \times 2 = 40$$

Option Class

All options of the same type on a stock.

Examples:

- IBM Calls
- IBM Puts

Each represents an option class.

Option Series

Options having:

- Same underlying asset
- Same strike price
- Same expiration date

Example:

IBM October 200 Calls

This represents one option series.

Moneyness of Options

The relationship between stock price and strike price determines whether an option is:

- In the Money (ITM)
 - At the Money (ATM)
 - Out of the Money (OTM)
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Call Options

Let:

$$S = \textit{StockPrice}$$

$$K = \textit{StrikePrice}$$

In the Money

$$S > K$$

Exercise produces immediate profit.

At the Money

$$S = K$$

No immediate gain from exercise.

Out of the Money

$$S < K$$

Exercise would be irrational.

Put Options

In the Money

$$S < K$$

At the Money

$$S = K$$

Out of the Money

$$S < K$$

Intrinsic Value

Intrinsic value is:

The value of an option if it were exercised immediately.

Call Option

$$\max(S - K, 0)$$

Put Option

$$\max(K - S, 0)$$

Time Value

An option often trades above intrinsic value.

Why?

Because there is still time remaining for favorable price movements.

Relationship

$$\text{Option Value} = \text{Intrinsic Value} + \text{Time Value}$$

Important Exam Insight

An in-the-money American option is worth at least its intrinsic value because:

The holder can exercise immediately.

However, early exercise is often not optimal because doing so destroys time value.

FLEX Options

FLEX stands for:

Flexible Exchange Options

These allow customized terms.

Possible customizations include:

- Strike Price
- Expiration Date
- Exercise Style

Example:

A European option may be created on an exchange that normally trades American options.

Why FLEX Options Exist

To compete with OTC markets.

They provide flexibility while still benefiting from exchange clearing and reduced credit risk.

Other Nonstandard Products

1. ETF Options

Options on Exchange Traded Funds.

Examples:

- SPY
- QQQ

They behave similarly to stock options.

2. Weeklys

Short-term options.

Characteristics:

- Created on Thursday
- Expire the following Friday

Useful for short-term trading strategies.

3. Binary Options

Provide a fixed payoff.

Example:

Call Strike = \$50

Payoff:

if stock finishes above \$50.

Otherwise:

0

Unlike standard options, payoff does not increase further after crossing strike.

4. CEBOs

Credit Event Binary Options.

Pay fixed amount if:

- Bankruptcy
- Failure to pay debt
- Debt restructuring

occurs before maturity.

These resemble credit default swaps.

5. DOOM Options

Deep-Out-of-the-Money Put Options.

Characteristics:

- Very cheap
- Very low strike price
- Pay only during market crashes

Used as catastrophe insurance.

Dividends and Stock Splits

Corporate actions can affect option values.

Cash Dividends

Most exchange-traded options are:

Not adjusted for normal cash dividends.

The strike price remains unchanged.

Large Dividends

Sometimes special adjustments occur.

If dividend is unusually large, exchanges may modify contract terms to preserve fairness.

Stock Splits

A stock split increases the number of shares outstanding.

Example:

2-for-1 split

Each old share becomes two shares.

Economic Effect

Total company value remains unchanged.

Therefore:

Stock price approximately halves.

Option Adjustment Rule

For an:

$$n : m$$

stock split:

New Strike Price

$$K_{new} = K_{old} \times \frac{m}{n}$$

New Number of Shares

$$Shares_{new} = Shares_{old} \times \frac{n}{m}$$

Example

Original Contract:

- Buy 100 shares
- Strike = \$30

Company announces:

2-for-1 split

Adjusted Contract:

- Buy 200 shares
- Strike = \$15

Economic position remains unchanged.

Stock Dividends

Stock dividends also increase shares outstanding.

Example:

20% stock dividend.

Investor receives:

1 new share for every 5 shares owned.

Option contracts are adjusted similarly to stock splits.

10.5 Trading

Historically:

Options were traded through open outcry trading floors.

Today:

Most trading is electronic.

Market Makers

Most option exchanges use market makers.

Role of Market Maker

A market maker continuously quotes:

Bid Price

Price willing to buy.

Ask (Offer) Price

Price willing to sell.

Bid-Ask Spread

$$Spread = Ask - Bid$$

This is a major source of profit for market makers.

Why Market Makers Matter

They provide:

Liquidity

Traders can always buy or sell.

Faster Execution

Orders are immediately executable.

Better Market Functioning

Prices remain continuously available.

Example

Suppose:

Bid:

4.90

Ask:

5.10

Spread:

0.20

Market maker earns spread while managing risk.

Offsetting Orders

Most option positions are closed before expiration.

Long Position

To close:

Sell identical option.

Short Position

To close:

Buy identical option.

Open Interest

Open interest measures:

Number of outstanding contracts.

Case 1

Neither trader closes a position.

Result:

Open Interest Increases

Case 2

One trader closes.

Result:

Open Interest Unchanged

Case 3

Both traders close positions.

Result:

Open Interest Decreases

Importance of Open Interest

High open interest generally indicates:

- Active trading
 - Greater liquidity
 - Easier entry and exit
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10.6 Commissions

Trading options involves commissions.

These vary across brokers.

Types of Brokers

Full-Service Brokers

Provide:

- Research
- Advice
- Portfolio services

Higher commissions.

Discount Brokers

Provide execution only.

Lower commissions.

Commission Structure

Often:

Fixed Fee + Percentage of Trade Value

Example

Suppose:

- Option Price = \$3
- Contracts = 8

Total trade value:

$$8 \times 100 \times 3 = 2400$$

Under Hull's sample schedule:

Commission:

$$\begin{aligned} 20 + 0.02(2400) \\ = 68 \end{aligned}$$

Closing an Option Position

If position is closed through an offsetting trade:

Another commission must be paid.

Exercising an Option

If option is exercised:

Commission resembles the commission for buying or selling the underlying stock.

Why Commissions Matter

Many strategies appear profitable before commissions.

After commissions:

Profitability may disappear.

This is especially important for:

- Frequent trading
 - Short-term strategies
 - Small option positions
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Section Summary

Expiration Cycles

- January Cycle
 - February Cycle
 - March Cycle
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LEAPS

Long-term options with maturities up to 39 months.

Moneyness

Call:

- $ITM \rightarrow S > K$
- $ATM \rightarrow S = K$
- $OTM \rightarrow S < K$

Put:

- $ITM \rightarrow S < K$
 - $ATM \rightarrow S = K$
 - $OTM \rightarrow S > K$
-

Intrinsic Value

Call:

$$\max(S - K, 0)$$

Put:

$$\max(K - S, 0)$$

Option Value

$$\textit{Intrinsic Value} + \textit{Time Value}$$

Market Maker

Provides:

- Bid Price
 - Ask Price
 - Liquidity
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Open Interest

Measures outstanding contracts.

FLEX Options

Customized exchange-traded options.

Important Nonstandard Products

- ETF Options
 - Weeklys
 - Binary Options
 - CEBOs
 - DOOM Options
-

Quick Revision Sheet

Standard Contract Size

100 shares

LEAPS

Long-term options up to 39 months

Call ITM

$$S > K$$

Put ITM

$$S < K$$

Intrinsic Value of Call

$$\max(S - K, 0)$$

Intrinsic Value of Put

$$\max(K - S, 0)$$

Option Value

$$\textit{Intrinsic} + \textit{Time}$$

Market Maker Profit

Bid-Ask Spread

Open Interest

Number of outstanding contracts

FLEX Options

Customized exchange-traded options

Weeklys

Created Thursday, expire next Friday

Binary Option Payoff

Fixed amount if strike condition met

Stock Split Adjustment

Strike falls, shares increase

End of Part 2 (Sections 10.4–10.6)